

Microservices Interview Questions (40)

1. What is a Microservice?
2. What are the advantages of Microservices over a Monolithic architecture?
3. What are the disadvantages of Microservice architecture?
4. When should you avoid using Microservices?
5. What are the core design principles of Microservices?
6. What is the 12-Factor methodology?
7. Why are Microservices typically designed to be stateless?
8. Why are Spring Boot and Spring Cloud commonly used for Microservices?
9. Should multiple Microservices share a database? Why or why not?
10. What is Distributed Tracing and why is it needed in Microservices?
11. How is Distributed Tracing implemented in Microservices?
12. How do Microservices communicate internally and externally?
13. What Microservice design patterns are commonly used and why?
14. What are common database design patterns in Microservices architecture?
15. What is the SAGA pattern in Microservices?
16. What is CQRS (Command Query Responsibility Segregation)?
17. What architecture patterns are best for read-heavy vs write-heavy systems?
18. What is the difference between Choreography and Orchestration in Microservices?
19. What fault tolerance mechanisms are commonly used in Spring Microservices?
20. What is the Circuit Breaker pattern and how does it work?
21. Which Spring Boot annotations are used for implementing Circuit Breakers?
22. What libraries are commonly used for Circuit Breaker implementation?
23. How can methods be executed asynchronously in Spring Boot?
24. How can one Microservice call another asynchronously?
25. What communication styles exist between Microservices?
26. How can you restrict one Microservice from calling another?
27. How should credentials be securely stored in Spring Boot Microservices?
28. What is Service Discovery and why is it required?
29. How is configuration management handled in Microservices?
30. How do you version Microservice APIs?
31. What is an API Gateway and why is it used in Microservices architecture?
32. What is the role of a Service Mesh in Microservices?

33. What are Idempotent APIs and why are they important in distributed systems?
34. How do you handle data consistency across multiple Microservices?
35. What is Event-Driven Architecture in Microservices?
36. What is Event Sourcing and how does it work?
37. How do you implement rate limiting in Microservices?
38. What is the Bulkhead pattern in Microservices resilience design?
39. How do you monitor and observe Microservices in production?
40. What challenges arise when testing Microservices and how can they be solved?